■ Building Your First Full-Stack Mini App!

Capstone Kickoff Session | Detailed Lesson Plan for Educators

Grade Level	Grades 4 – 7	**Duration**	60 – 75 minutes
Subject	Computer Science / Coding	**Format**	Interactive Lecture + Activity
Prerequisites	Basic HTML/CSS awareness	**Materials**	Paper, pencils, coloured pens

■ Learning Outcomes

By the end of this session, students will be able to:

- ✔ Define **Full-Stack** development and distinguish front-end from back-end
- ✔ Identify front-end and back-end components in familiar apps (Swiggy, YouTube)
- ✔ Choose a mini-app concept and articulate it as a **User Story**
- ✔ Sketch a basic **wireframe (Naksha)** labelling FE/BE parts
- ✔ Present their app idea confidently to peers

Section 1: Lesson Overview & Timing

This session introduces students to the concept of full-stack development through India-centric real-world examples (Swiggy, Haldirams, Dabbawalas). Students then plan their own Capstone mini-app. The session is designed to be high-energy, interactive, and creative, balancing instruction with hands-on activity.

Time	Phase	Activity	Mode
0 – 5 min	Warm-Up	Raise Your Hand! (Swiggy/YouTube poll)	Whole Class
5 – 15 min	Concept 1	What is Full-Stack? FE vs BE explained	Direct Instruction
15 – 25 min	Real-World Tour	Swiggy, Dabbawala & Haldiram examples	Discussion
25 – 28 min	Quick Quiz	Order Now button — FE or BE?	Interactive
28 – 38 min	Concept 2	Mini Apps = Chai Stall philosophy	Direct Instruction
38 – 55 min	App Naksha	Choose app, write user story, wireframe	Individual Work
55 – 65 min	Share & Celebrate	Peer presentation + Badge Award	Group Share
65 – 75 min	Wrap-Up	Homework brief + Next session preview	Whole Class

Section 2: Detailed Teaching Notes

2.1 Warm-Up: Did You Know? (0 – 5 min)

Purpose: Activate prior knowledge and build excitement. Establish that apps students already love are built by people who started just like them.

Educator Script:

> *"Raise your hand if you have used Swiggy to order food! Keep it up if you have watched YouTube today. Amazing — behind every single one of those apps are thousands of lines of code. But guess what? It ALL started with one idea, one person thinking: I can build this. Today, YOU are going to build your very own mini app!"*

Engagement tips:

- Count hands aloud — make it theatrical ("Wow, almost everyone!").
- Ask one student: 'What is your favourite app and why?' (30-second share).
- Display the first slide on the projector while students settle.

2.2 Concept 1: What is Full-Stack? (5 – 15 min)

Introduce the two halves of every digital product. Use simple, concrete language. Avoid technical jargon — anchor every explanation to something students can see and touch.

Aspect	■ Front-End	■■ Back-End
What it is	Everything you SEE on screen	Everything working invisibly behind the scenes
Who interacts	The user / student	Servers, databases, code engines
Example (Swiggy)	The menu, buttons, images, score	Saving your address, processing payment
Colour code	ORANGE in your wireframe	BLUE in your wireframe
Analogy	The front of a chai stall — display	The kitchen making the chai

Checking for understanding: Ask the class: 'If I tap the red heart icon to like a video on YouTube, is that front-end or back-end?' (Answer: the icon is FE; storing the like in the database is BE — both happen together!)

2.3 Real-World Tour: India App Examples (15 – 25 min)

Walk through three culturally familiar examples. Encourage students to call out front-end and back-end features before you reveal them.

■ Swiggy

FE: Menu display, 'Order Now' button, delivery tracker animation, restaurant photos.

BE: User authentication, address storage, restaurant inventory, payment processing, GPS tracking logic, notification triggers.

■ Dabbawala App

FE: Live map showing your lunchbox en route, estimated arrival time, order status badges.

BE: Route optimisation algorithm, hub-tracking database, delivery partner assignment, SMS/push notification system.

■ Haldiram's

FE: The visual menu — photos of gulab jamun, samosas, prices, category filters.

BE: Inventory management (how many gulab jamuns left), order processing pipeline, billing system, daily sales analytics.

Section 2 (continued): Detailed Teaching Notes

2.4 Quick Quiz: Front-End or Back-End? (25 – 28 min)

Fast-paced formative check. Read each item aloud; students point LEFT for FE, RIGHT for BE. Keep it playful — celebrate correct answers, correct misconceptions gently.

Feature / Action	FE or BE?	Why
The 'Order Now' button on Swiggy	■ **Front-End**	Visible UI element the user taps
Saving your delivery address	■■ **Back-End**	Stored in a server database
The star-rating display on a restaurant	■ **Front-End**	Visual component on screen
Calculating your total bill with discount	■■ **Back-End**	Logic runs on the server
The animated loading spinner	■ **Front-End**	CSS/JS animation users see
Sending you an OTP on your phone	■■ **Back-End**	Server-side SMS/API trigger
The search bar where you type restaurant names	■ **Front-End**	Input element on the UI
Remembering your past orders	■■ **Back-End**	Database query & storage

2.5 Concept 2: Mini Apps Are Like Chai Stalls (28 – 38 min)

Introduce the concept of a **Capstone project** — a single focused mini-app that does one thing perfectly. Use the chai stall analogy: a great chai stall masters tea, not pizza, not biryani. Scope is a feature, not a limitation.

Key teaching points:

- A mini-app has **one clear purpose** — quiz, reminder, recipe book, etc.
- Every great product (Swiggy, Zomato, IRCTC) started as a tiny idea with limited features.
- The Capstone is their **final project** — where all their HTML, CSS, and JS skills come together.
- Encourage creativity: they can choose from the suggested options OR invent something entirely new.

App Options to Present:

Option	App Name	Core Feature	Good For Students Who Like...
A	India Quiz App	Multiple-choice Q&A; with scoring	General knowledge, geography, cricket

B	Family Recipe Book	Add & view recipes with ingredients	Food, cooking, family traditions
C	Class Birthday Reminder	Enter names & dates, see countdown	Organisation, friends, celebrations
D	Your Own Idea!	Completely custom concept	Creative thinkers, innovators

Section 3: App Naksha (Blueprint) Activity — 38 to 55 min

This is the core hands-on segment. Students work individually to plan their mini-app. Circulate and provide one-on-one guidance. The activity has four structured steps.

Step 1

Choose Your App

Students select from the four options (or propose their own). They write the app name in large letters at the top of their planning sheet. Encourage bold, fun names: 'CricketGuru', 'MaaKiRasoi', etc.

Step 2

Write Your User Story

Teach the user story format: 'I am [persona], I want [feature], so that [benefit].' Example: 'I am Priya, a Grade 4 student. I want a quiz app testing state capitals, so that I can score better in Geography exams!' Students write 2–3 sentences.

Step 3

Name Your App

Students brainstorm and choose a memorable, creative name. Encourage a mix of Hindi + English (e.g., 'QuizGuru', 'JanmadnApp', 'RecipeWala'). The name should reflect what the app does.

Step 4

Draw Your Wireframe

Students sketch a simple phone-shaped box and draw the key screens: Home screen, main feature screen, and result/output screen. They then label each element ORANGE (front-end) or BLUE (back-end). Even stick figures and rough boxes count!

■ **Educator Tip:** If students are stuck, ask guiding questions: 'Who will use your app? What problem does it solve? What is the ONE thing it must do?' Remind them that a simple idea done well is better than a complicated idea half-done.

Section 4: User Story Writing Guide

User stories are a professional technique used by real software teams. Teaching students this format builds both writing skills and computational thinking. Here is a complete guide for the classroom.

The Formula:

"I am [WHO I am], I want [WHAT I want the app to do], so that [WHY / the benefit I get]."

Sample User Stories:

App	User Story Example
India Quiz App	I am Arjun, a Grade 5 student who loves cricket. I want a quiz with questions about Indian cricketers and famous stadiums, so that I can impress my friends with my knowledge.
Family Recipe Book	I am Meera, a Grade 6 student. I want a place to save my grandmother's recipes with ingredient lists, so that our family recipes are never lost.
Birthday Reminder	I am Ravi, a Grade 4 student with 40 classmates. I want to enter everyone's birthday and see who is celebrating soon, so that I never forget to wish a friend.
Custom App	I am [your name], a [grade] student who loves [interest]. I want [your idea], so that [your reason].

Section 5: Wireframe (Naksha) Drawing Guide

A wireframe is a rough sketch of an app screen — no art skills needed! Students draw boxes and labels to show WHERE things will appear on the screen. This is exactly what professional UI/UX designers do before any code is written.

What to Include in the Wireframe:

■ **App Name / Logo area**	A box at the top for the app title — every app has this!
■ **Input Fields**	Boxes where users type information (name, date, search term).
■ **Buttons**	Rounded rectangles labelled with action words: Submit, Save, Next, Quiz Me!
■ **Display Area**	Where results / content show up — a list of recipes, quiz score, countdown.
■ **Navigation**	Optional: home icon, back arrow, tab bar at the bottom.

Colour Coding Instructions:

■ **ORANGE**	= Front-End elements — anything the user SEES and TOUCHES: buttons, text, images, input boxes, animations.
■ **BLUE**	= Back-End elements — anything that stores or processes data behind the scenes: saving data, checking login, calculating score.

Section 6: Differentiation & Classroom Management

■ For Beginners	■ For Advanced Students
• Provide a printed wireframe template with phone outline already drawn. • Offer sentence starters for the user story. • Limit wireframe to one screen only. • Pair with a buddy for the app selection step.	• Plan THREE screens (home, feature, results). • Write TWO different user stories for two personas. • Add a 'stretch' feature to their app idea. • Draft a list of questions they would need to answer in code.

Common Misconceptions to Address:

Misconception: 'The whole app is front-end because I can see everything.'

Fix: Remind students: behind every screen there is a machine remembering, calculating, and storing. They can't see the kitchen from the dining table, but the food still gets made!

Misconception: 'My app idea is too simple.'

Fix: Validate simplicity! The best apps do one thing extremely well. Instagram started as just photo sharing. WhatsApp was just messaging.

Misconception: 'I can't draw the wireframe — I'm not artistic.'

Fix: Reassure students that wireframes are intentionally rough. Professional designers use squiggles and boxes. Skill in art is irrelevant here.

Section 7: Share & Celebrate (55 – 65 min)

This segment builds confidence through peer sharing. Every student deserves a moment to present their idea — even if it is just one sentence. Create a safe, celebratory atmosphere.

■ **Pair Share (2 min)**	Students turn to the person next to them and share: App name + one sentence about what it does.
■ **Whole Class (5 min)**	Invite 4–5 volunteers to share with the full class. Ask the class to snap fingers or clap after each share.
■ **Badge Award (2 min)**	Announce that every student who completed their Naksha has earned the 'App Architect Level 1' badge!
■ **Positive Feedback (1 min)**	Invite the class to shout one encouraging word to the presenters: Amazing! Brilliant! Superb!

Section 8: Homework & Next Session Preview

■ **Homework Task (5 – 10 minutes at home):**

Show your App Naksha to one family member (parent, sibling, grandparent). Ask them two questions:
1. **'Would you use this app? Why or why not?'**
2. **'What would you change or add?'**

Write one sentence of feedback you received in your coding journal or on the back of your Naksha sheet. Bring it to the next session!

Next Session Preview — Building the Front-End in HTML:

→ Students will open their code editor for the first time in this project.

→ They will convert their wireframe sketches into actual HTML structure (divs, headings, buttons, inputs).

→ Introduction to linking a CSS file for basic styling.

→ By end of Session 2, each student will have a visible, clickable front-end page!

Section 9: Assessment Rubric — App Naksha

Criteria	4 — Excellent	3 — Proficient	2 — Developing	1 — Beginning

App Concept Clarity	App idea is clear, original, and focused on one task	App idea is clear and focused	App idea is somewhat vague	App idea is unclear or missing
User Story	Complete user story with specific who/what/why	User story has all 3 parts	User story is incomplete	No user story written
Wireframe Quality	3+ screens sketched with labels and colour coding	1–2 screens with most elements labelled	Wireframe present but minimal labels	No wireframe or very incomplete
FE/BE Identification	All elements correctly colour-coded FE/BE	Most elements correctly labelled	Some labelling attempts made	No FE/BE labelling
Presentation	Shared idea clearly and confidently	Shared idea with some confidence	Shared briefly when prompted	Did not share

Section 10: Educator Resources & Vocabulary

Key Vocabulary to Reinforce:

Term	Definition
Full-Stack	A complete application that includes both what users see (front-end) and what works behind the scenes (back-end).
Front-End	The visual, interactive part of an app — HTML, CSS, JavaScript elements that appear on screen.
Back-End	Server-side logic, databases, and APIs that process and store data invisibly.
User Story	A short description of a feature from the user's perspective: who, what, and why.
Wireframe	A rough sketch or blueprint showing the layout of an app screen.
Capstone Project	A final, comprehensive project that demonstrates all the skills learned in a course.
Mini App	A focused application that does one specific task very well.
UI	User Interface — the visual elements a user interacts with.

Educator Checklist — Before the Session:

- ■ Print wireframe planning sheets (one per student) — a blank phone outline helps beginners.
- ■ Prepare orange and blue coloured pens/pencils for wireframe colour coding.
- ■ Load the Swiggy / YouTube / Haldiram's websites or app screenshots on the projector.
- ■ Print or display the 'App Architect Level 1' badge to award at the end.
- ■ Have blank paper available for students who want more space to sketch.

■ Test the quiz segment by reading questions aloud once to check timing.

■ You are now ready to run an amazing Full-Stack Mini App Kickoff session! Remember: the goal is inspiration, not perfection. Every great app builder started exactly here. ■